CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 5: Database Recovery Protocols & WAL Logging

Course: Advanced Database Engineering [23CS3503]
Category: Assignments | Assignment #5 | Academic Year: 2026-27

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1. Overview & Objective:

Failure models, Write-Ahead Logging WAL , checkpointing, and ARIES recovery algorithm.

2. Assignment Problems & Instructions:

1. Explain the Write-Ahead Logging WAL protocol and why Steal/No-Force buffer management is preferred.
2. Detail the three phases of ARIES recovery algorithm: Analysis, Redo, and Undo.
3. Trace compensation log records CLRs created during transaction rollback upon system crash.
4. Differentiate between non-fuzzy checkpoints and fuzzy checkpoints with dirty page tables.

Submission Guidelines & Verification:

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